

day 39 prep

I need to do fourier transforms after having begun talking about fourier series last time in class. we had to discuss more complex numbers and euler's formula than i had hoped to get some background material in, but now I hope that we are ready to dive in to the python version of these things.

last time we talked about a fourier series, but we didn't talk about why we would ever do that type of transform. in order to do that, I'm going to show you a plot and have you guess what it is made of.

first we will start off easy, with just a sine or cosine wave, but soon the process becomes very difficult to do.

so what do we do now that we know that spotting these things is hard to do?

```
In [31]: x0 = np.linspace(0,10, 500)
         y0 = np.sin(2*np.pi*x)
```

```
In [32]: x1 = x
         y1 = np.sin(2*np.pi*x)+np.sin(4*np.pi*x)
```

```
In [33]: x2 = x
         y2 = np.sin(2*x)+np.sin(5*x)+np.sin(10*x)
```

```
In [34]: plt.close()
```

```
In [35]: plt.plot(x0, y0, x1, y1, x2, y2)
```

```
Out[35]: [<matplotlib.lines.Line2D at 0x7253a4f06a80>,
          <matplotlib.lines.Line2D at 0x7253a57ea630>,
          <matplotlib.lines.Line2D at 0x7253a57ea2d0>]
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [36]: %store x0 y0 x1 y1 x2 y2
```

```
Stored 'x0' (ndarray)
Stored 'y0' (ndarray)
Stored 'x1' (ndarray)
Stored 'y1' (ndarray)
Stored 'x2' (ndarray)
Stored 'y2' (ndarray)
```

In []:

In []:

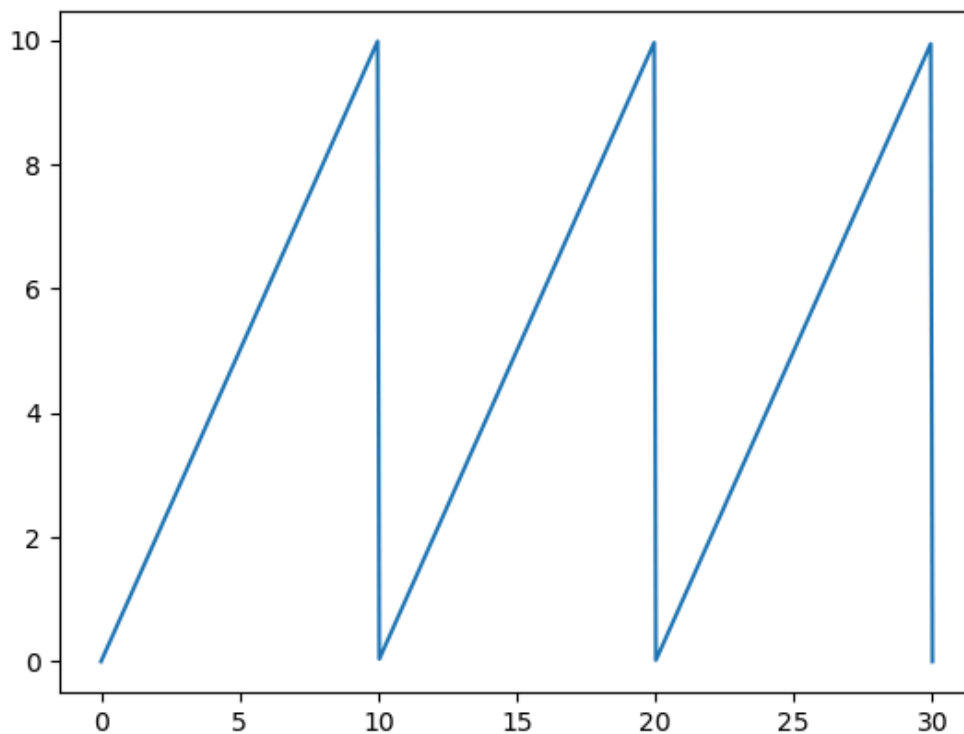
In []:

```
In [37]: L = 10
x = np.linspace(0, 3*L, 500) # show three periods bc it looks cooler
y = (x % L)                  # periodic sawtooth

fig0,ax0=plt.subplots()
ax0.plot(x, y)
```

Out[37]: [

Figure



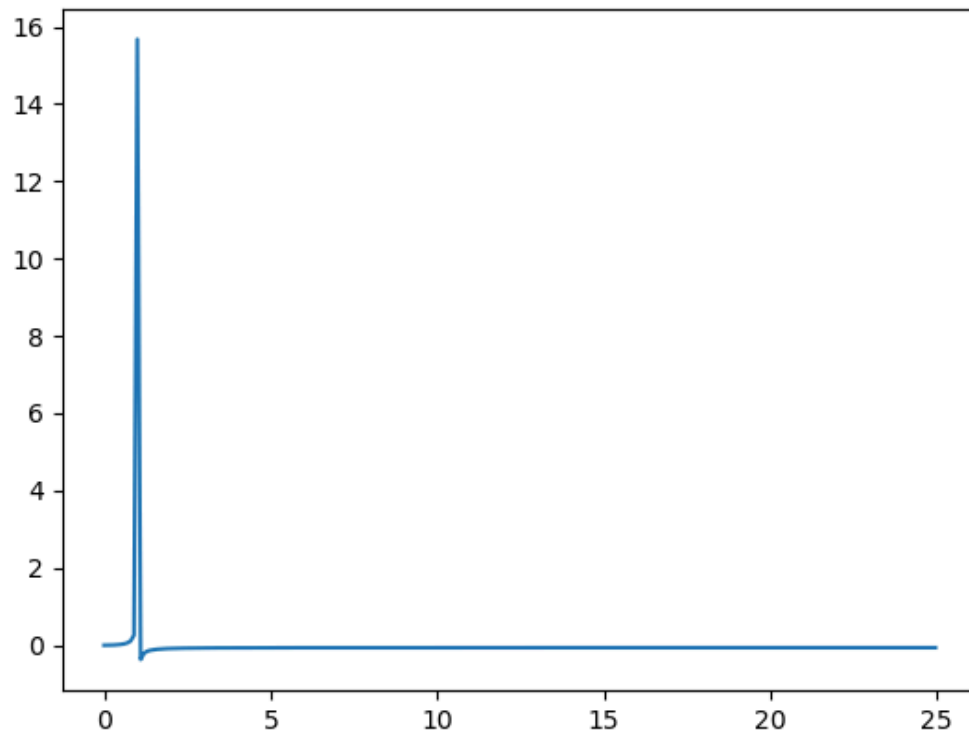
```
In [38]: c = np.fft.rfft(y0)
yback = np.fft.irfft(c)

w = np.fft.rfftfreq(500, d=x0[1]-x0[0])
```

```
In [39]: fig2, ax2 = plt.subplots()
ax2.plot(w, c.real)
```

Out[39]: [

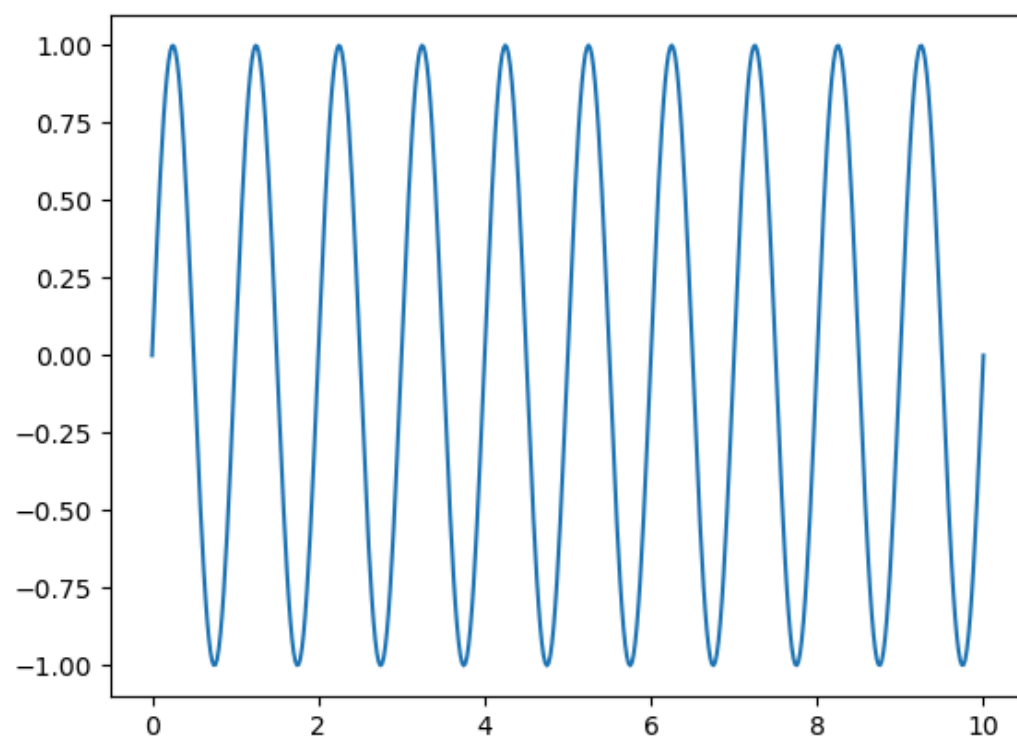
Figure



```
In [41]: fig4, ax4 = plt.subplots()
         ax4.plot(x0, yback)
```

```
Out[41]: [<matplotlib.lines.Line2D at 0x7253a4c85460>]
```

Figure



In []: